

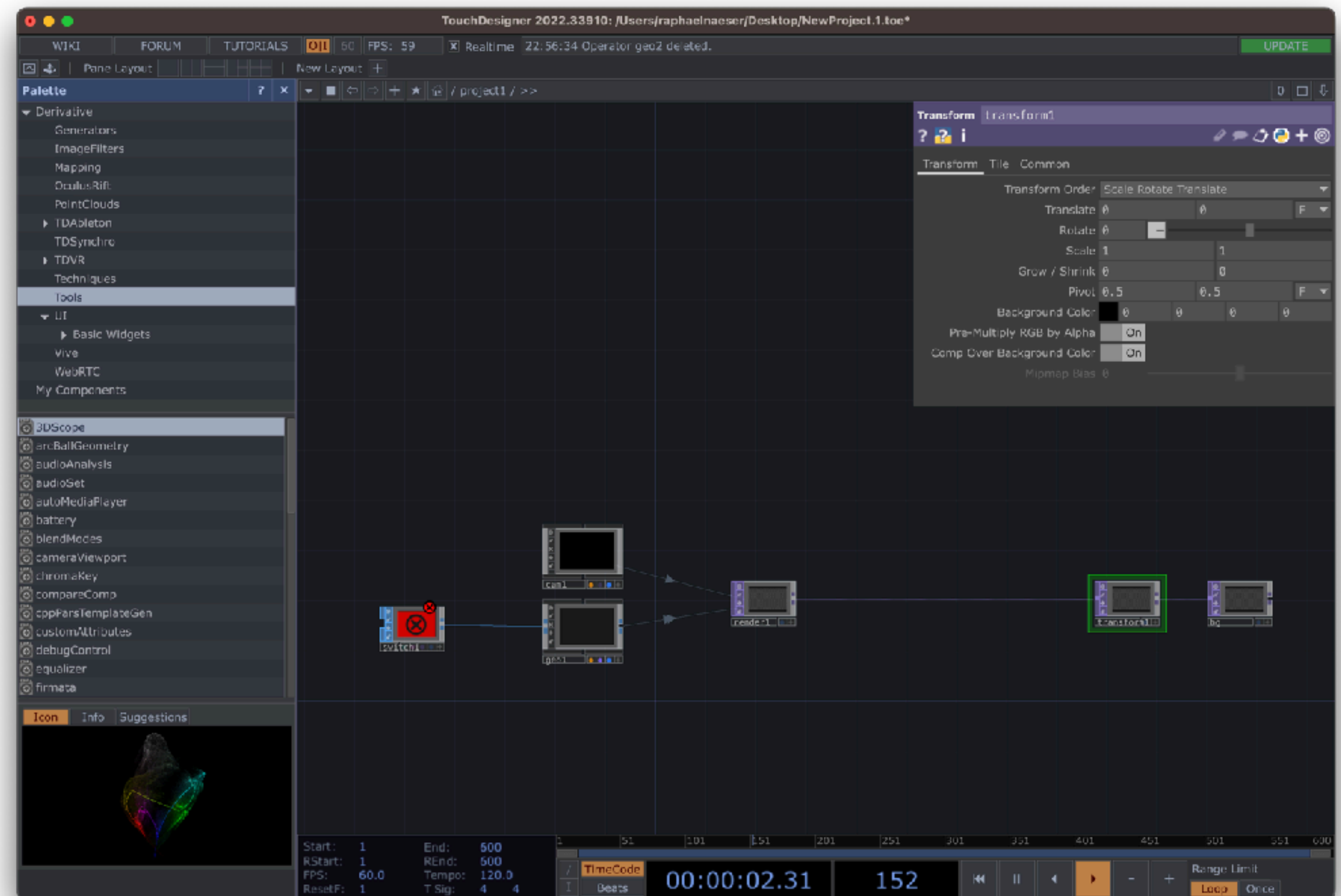
TD-Beginner 101 Workshop

V1.1

Drake Rubicon - 2024.07.17

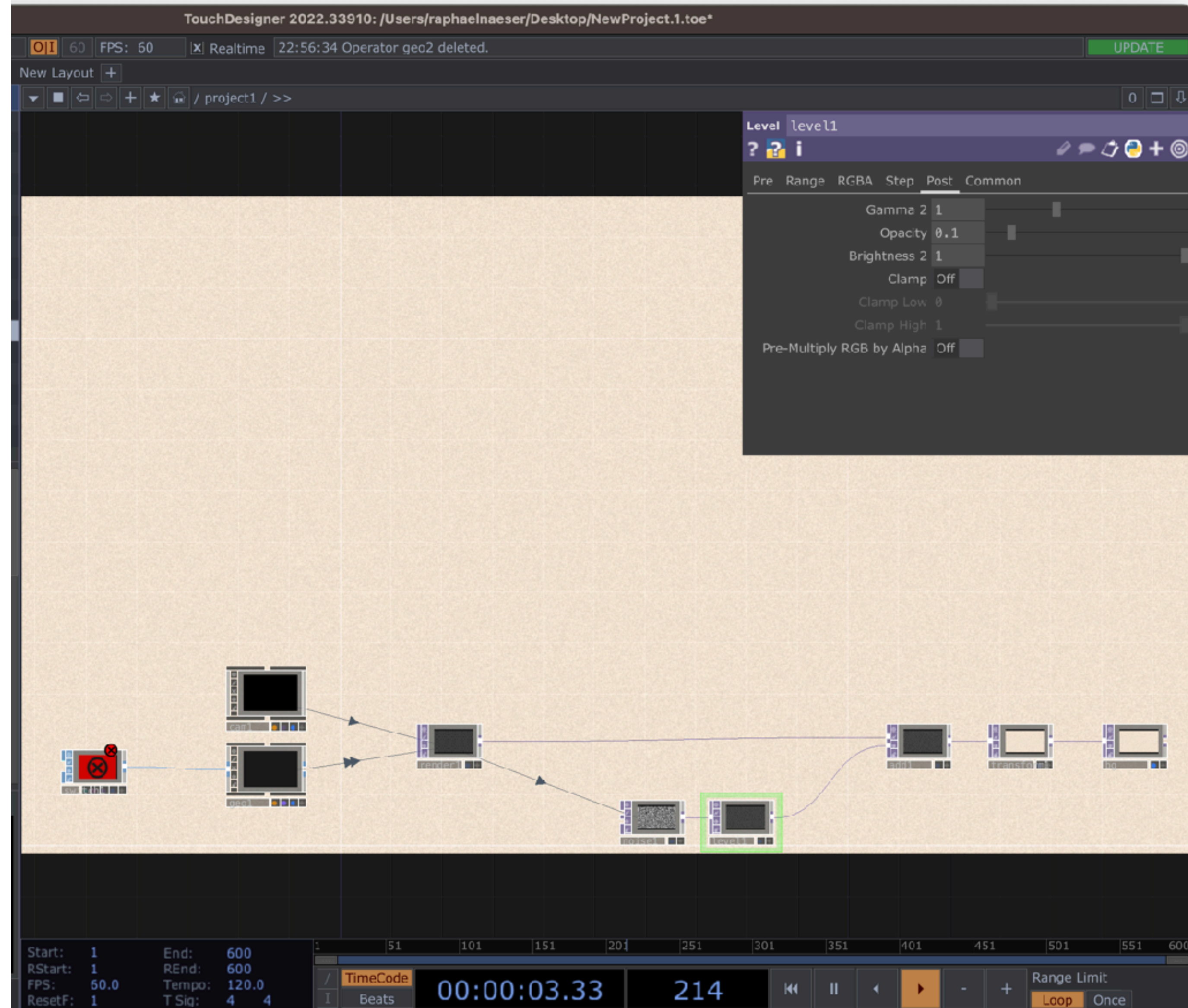
Part 1 - Handdrawn Lines

- Basic Render Setup
 - Cam, Geo, Render
 - Geo -> Standardkörper -> In/Out
 - Autogeo von Switch Sop
- Transform -> Comp over BG -> Hintergrund zu Papierfarbe
- Null -> "out1"
- Blauer Viewer-Punkt

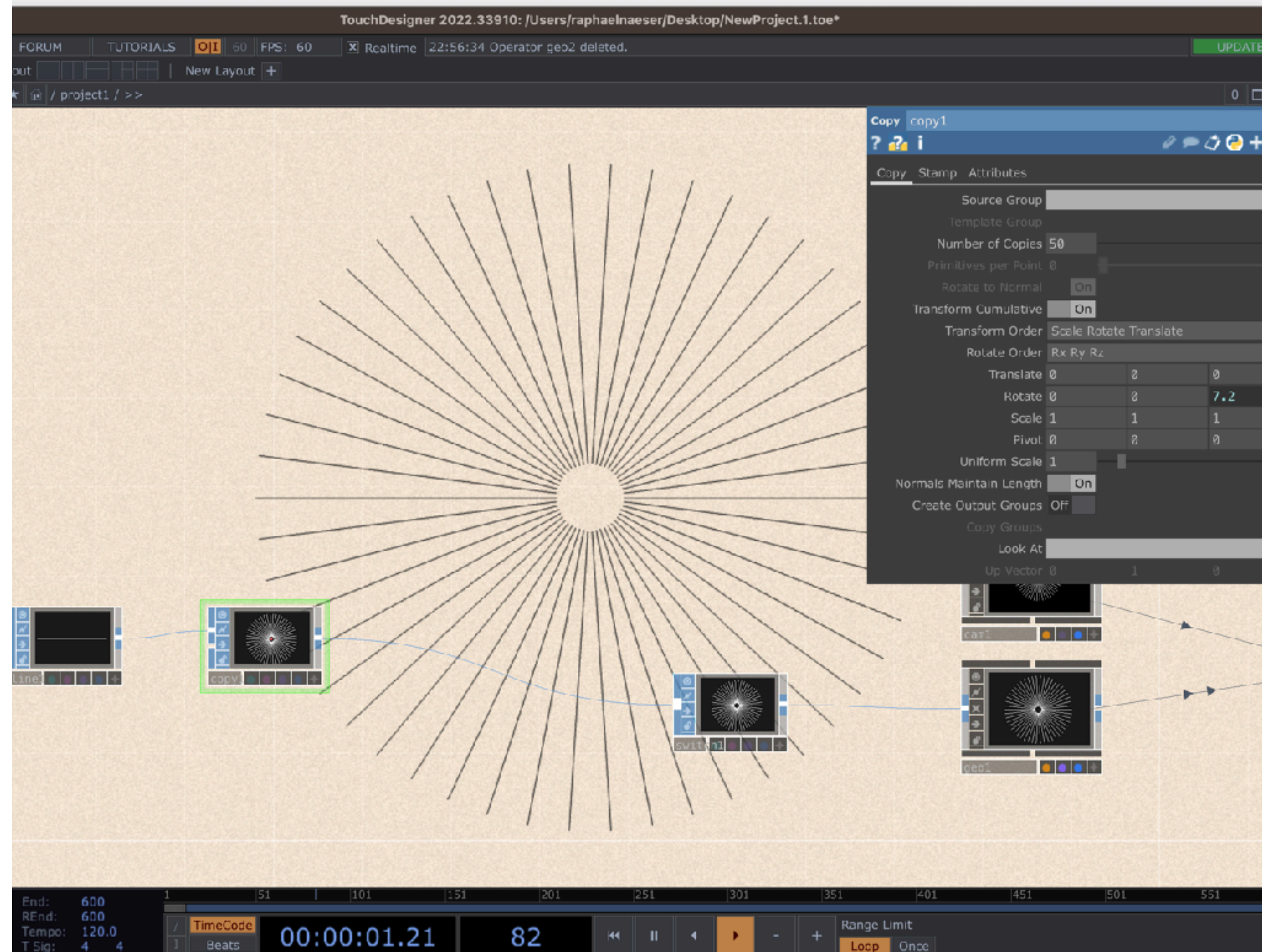


COMP - Components / TOP - Texture Operator

- Add auf Connection
- TOP Noise + Resolution:
 - `op('render1').width`
 - `op('render1').height`
 - Noise type random
- TOP Level
 - Post - Opacity 0.1

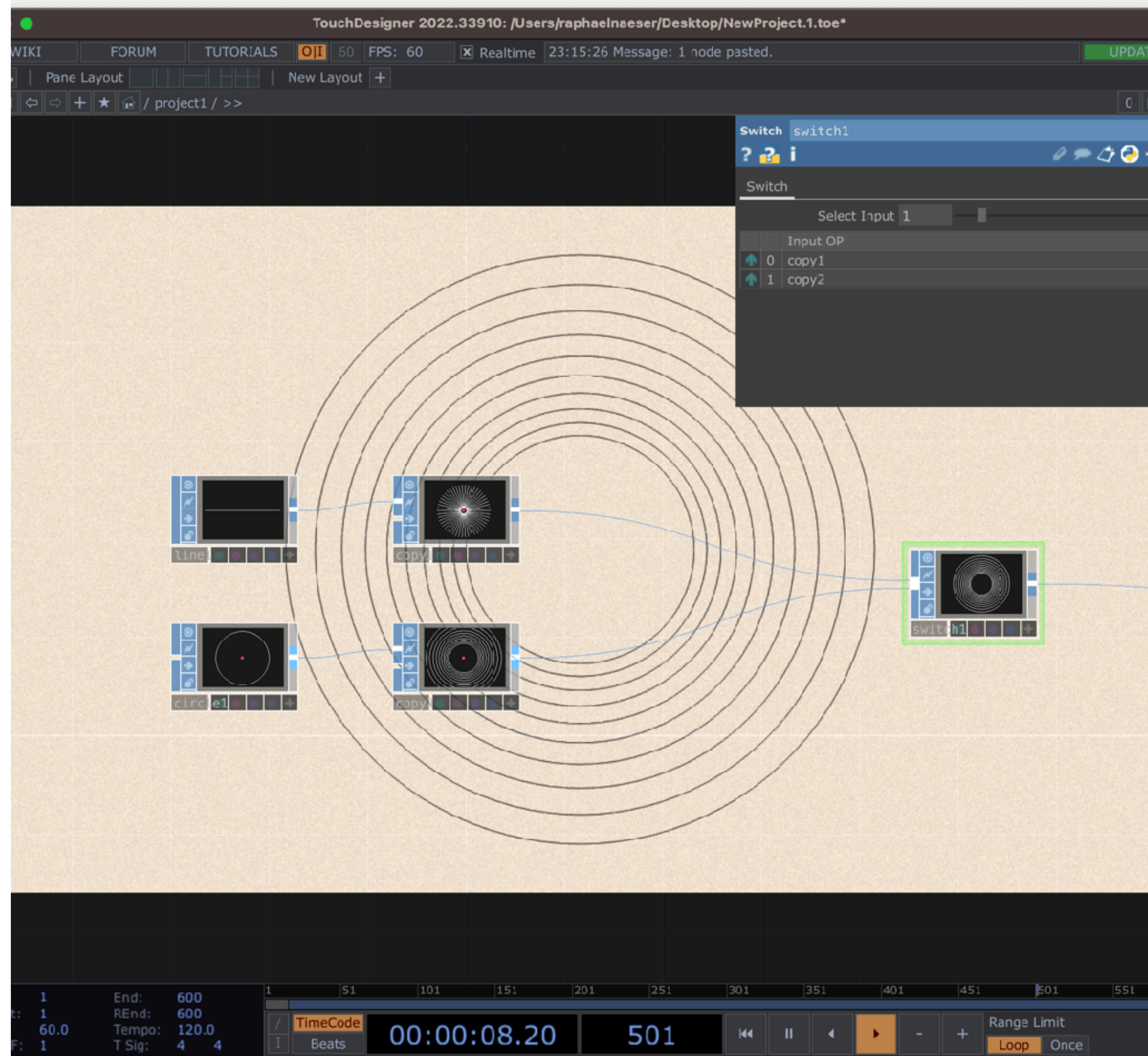


- SOP Line
 - Etwas verkürzen: Pa: .1 0 0
- SOP Copy
 - num of copies: 50
- Copy - Switch
- Ausrichten durch rotieren:
 - Rotate z: $360/\text{me.par.ncy}$
 - Hinweis: Hover = Parametername



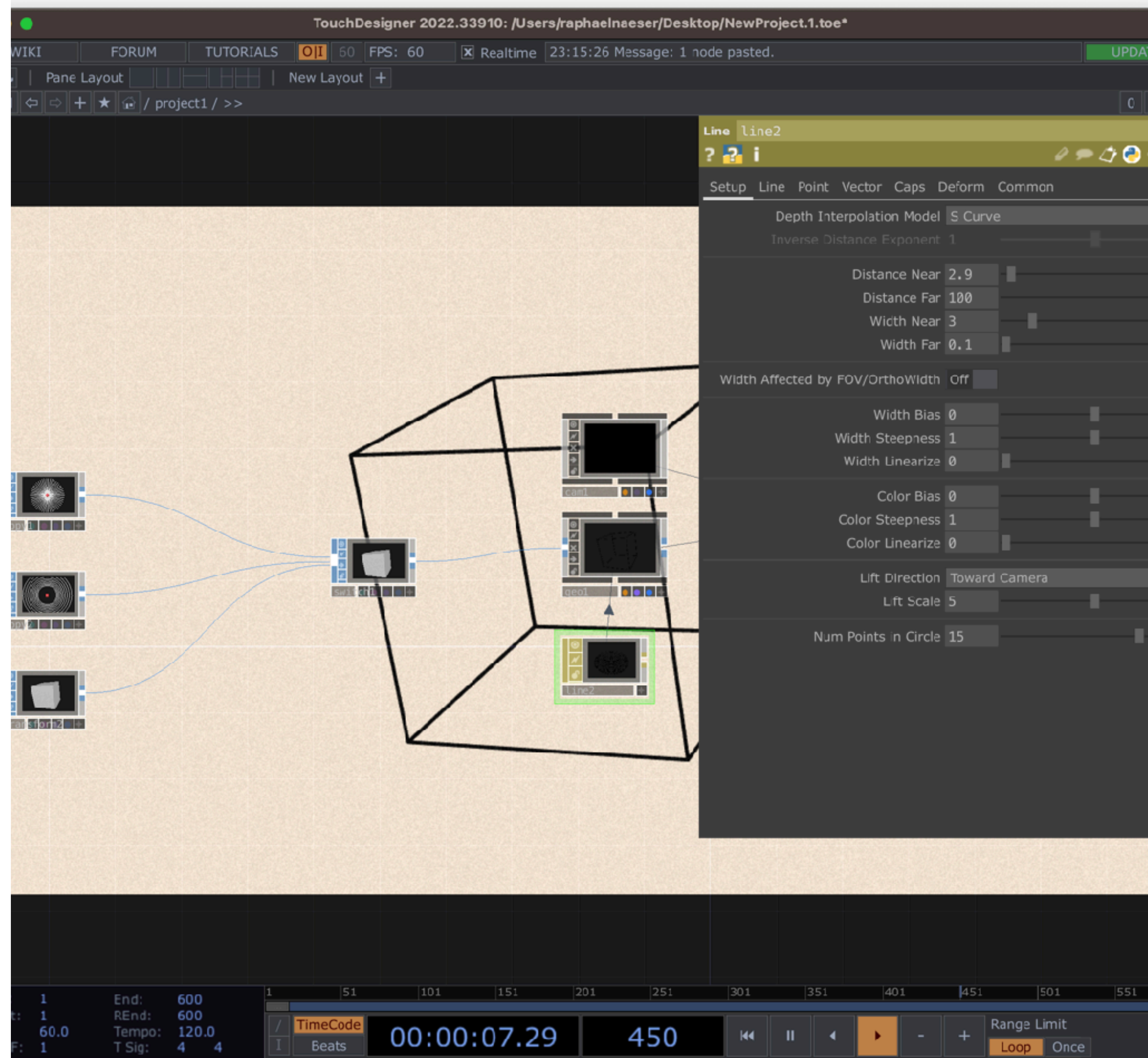
SOP - Surface Operator

- SOP Circle
 - Divisions: 100
 - Arc-Type: Open Arc
- SOP Copy
 - Uniform Scale 10
- Copy - Switch



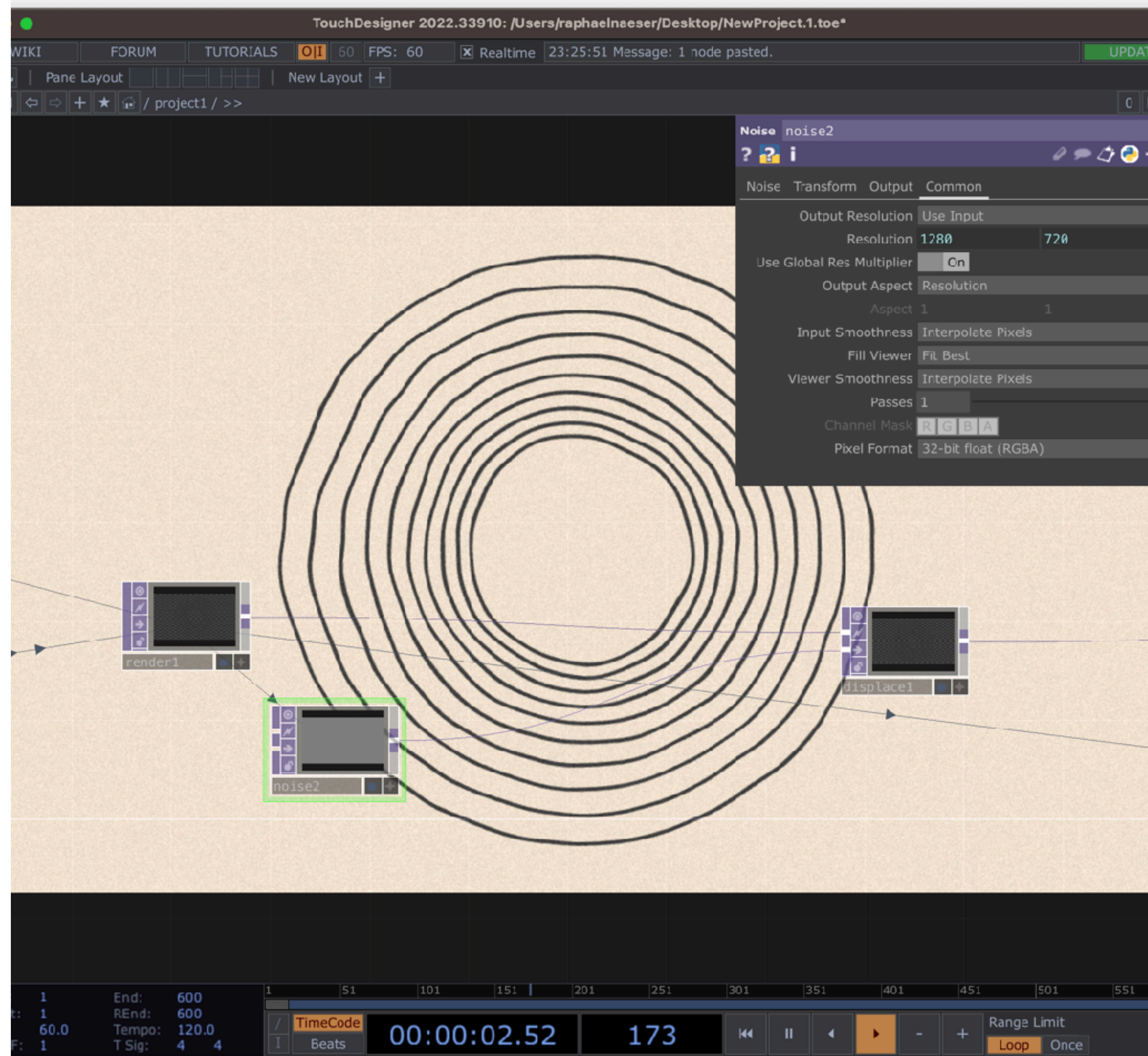
SOP - Surface Operator

- SOP Box
- SOP Transform
 - Rotation: 20 -25 0
- Vorschau im Switch: Volle Box
- 3D braucht Textur oder Material
- MAT Line
 - Auf Geo ziehen - Param Mat
 - Farbe wechseln Near/Far
 - Gleichm. Linien:
 - Dist. Far 100 // Width Near - Stärke



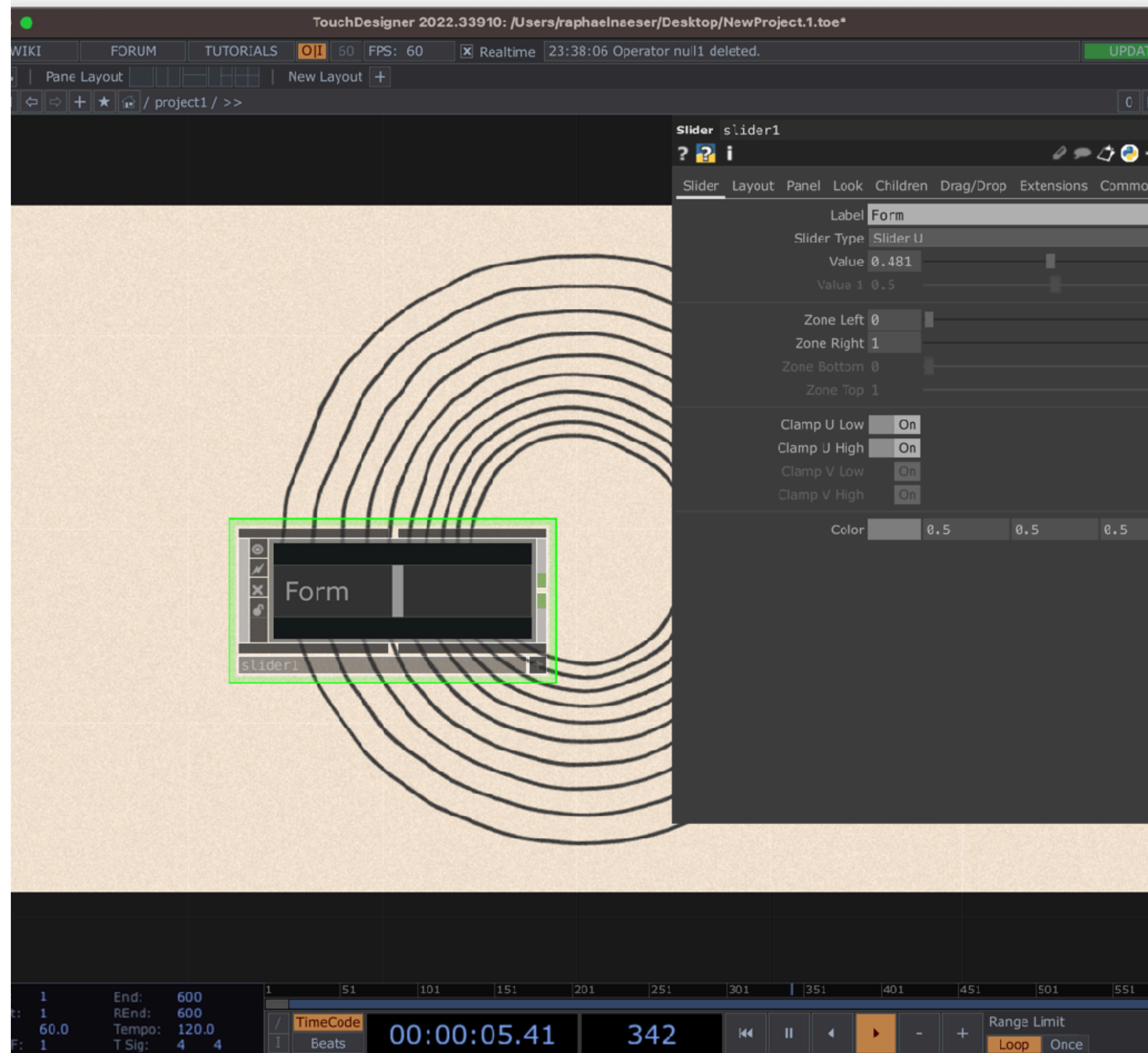
SOP - Surface Operator / MAT - Material

- TOP Displace
- Noise1 CopyPaste
- Verbinden
- Noise Type: Simplex 3D
- Monochrome: off
 - Amplitude 0.002
- Glitches beseitigen:
Common
Pixel Format 32bit float



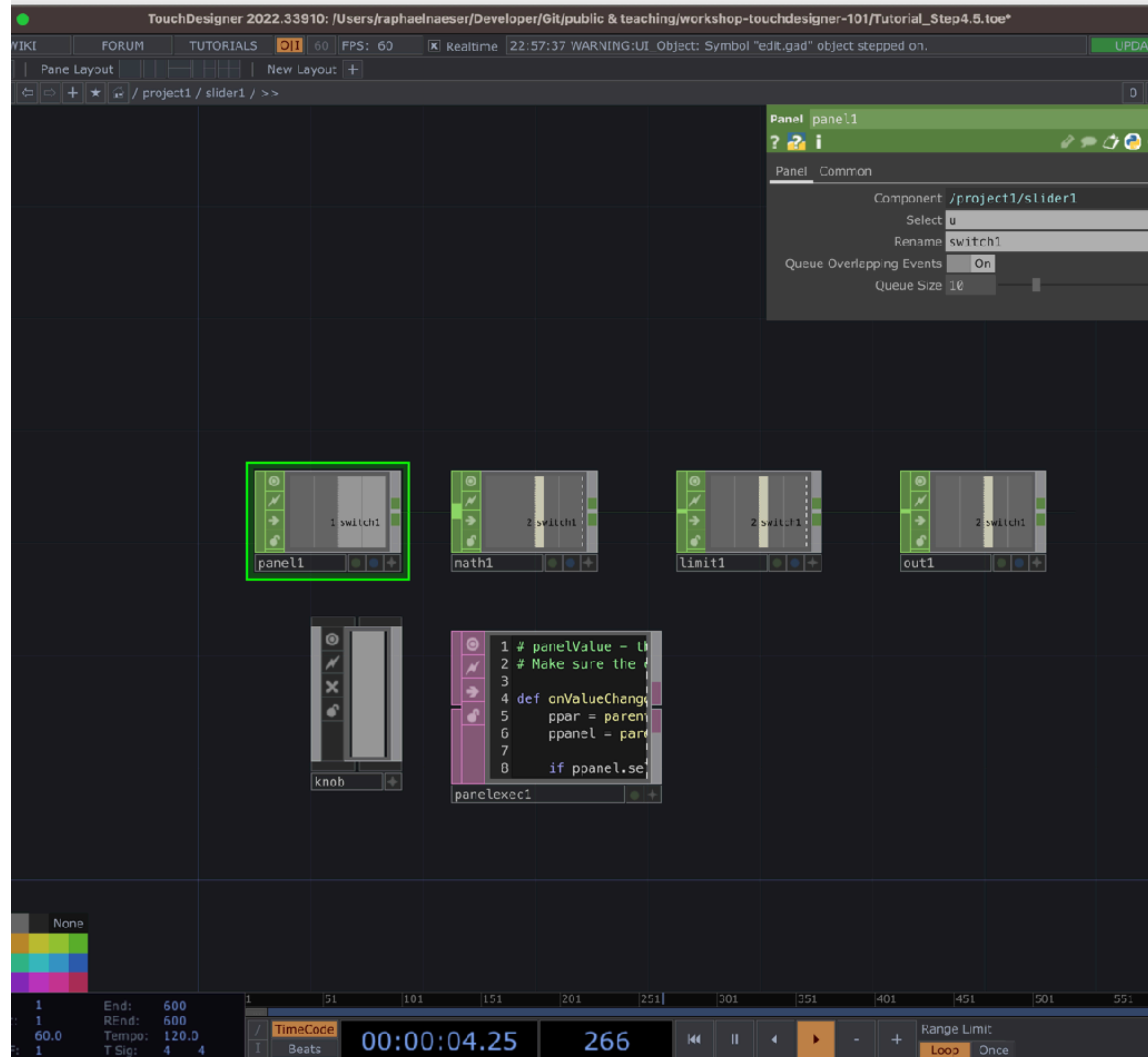
Part 2 - Interaktivität 1

- COMP Slider
 - Look: Farbe ändern
 - Name Geben
 - Autogeo von Switch Sop
- Viewer active -> Testen
- Slider geht von 0 bis 1
- 3 Formen -> 0, 1, 2
- Zoom in Slider

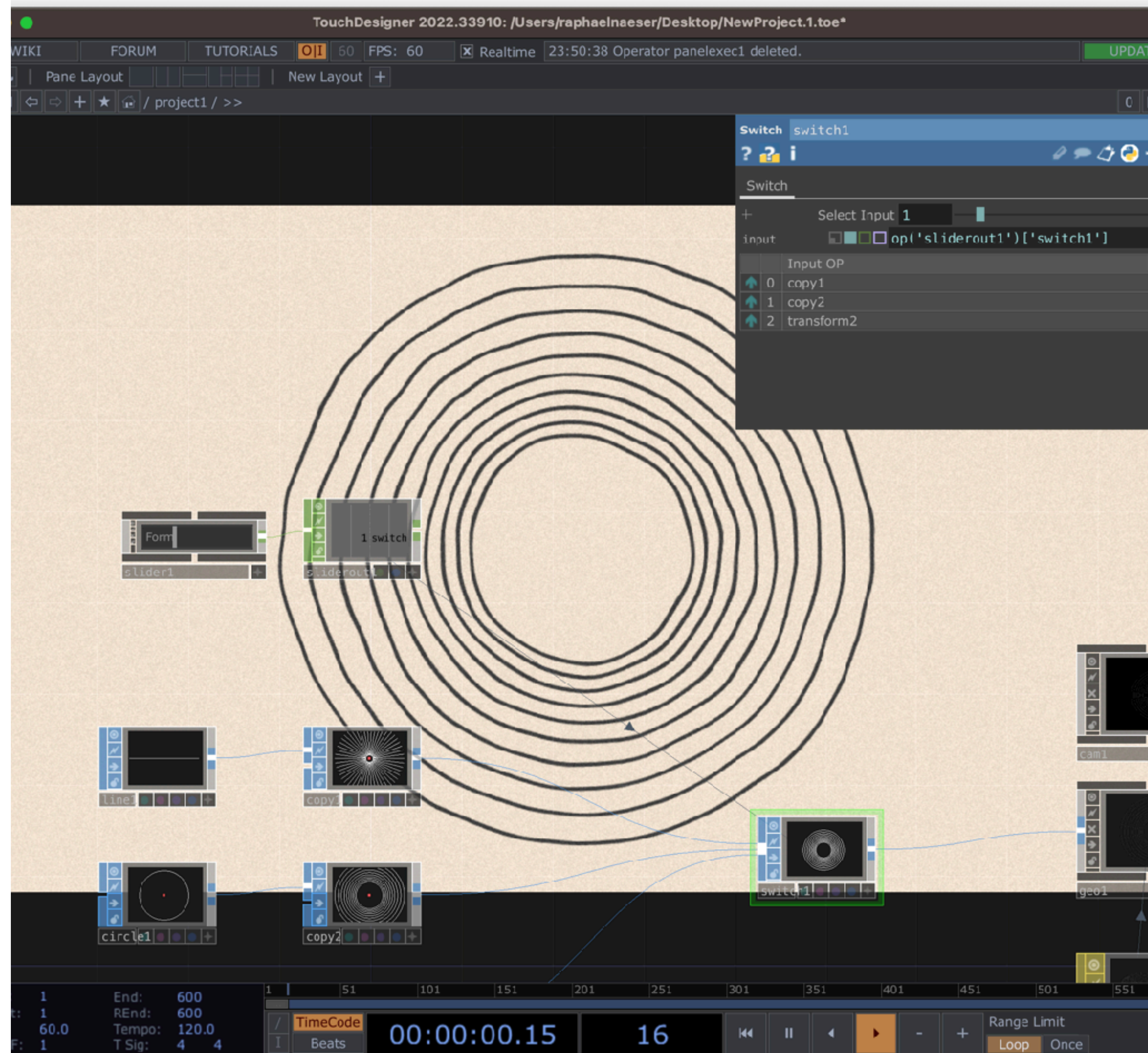


- par1 löschen
- CHOP panel1
 - select - u
 - Rename - switch1
- Out1 platz geben
- CHOP math1
 - Multiply 2
- CHOP Limit1
- Quantize Round
 - Step 1

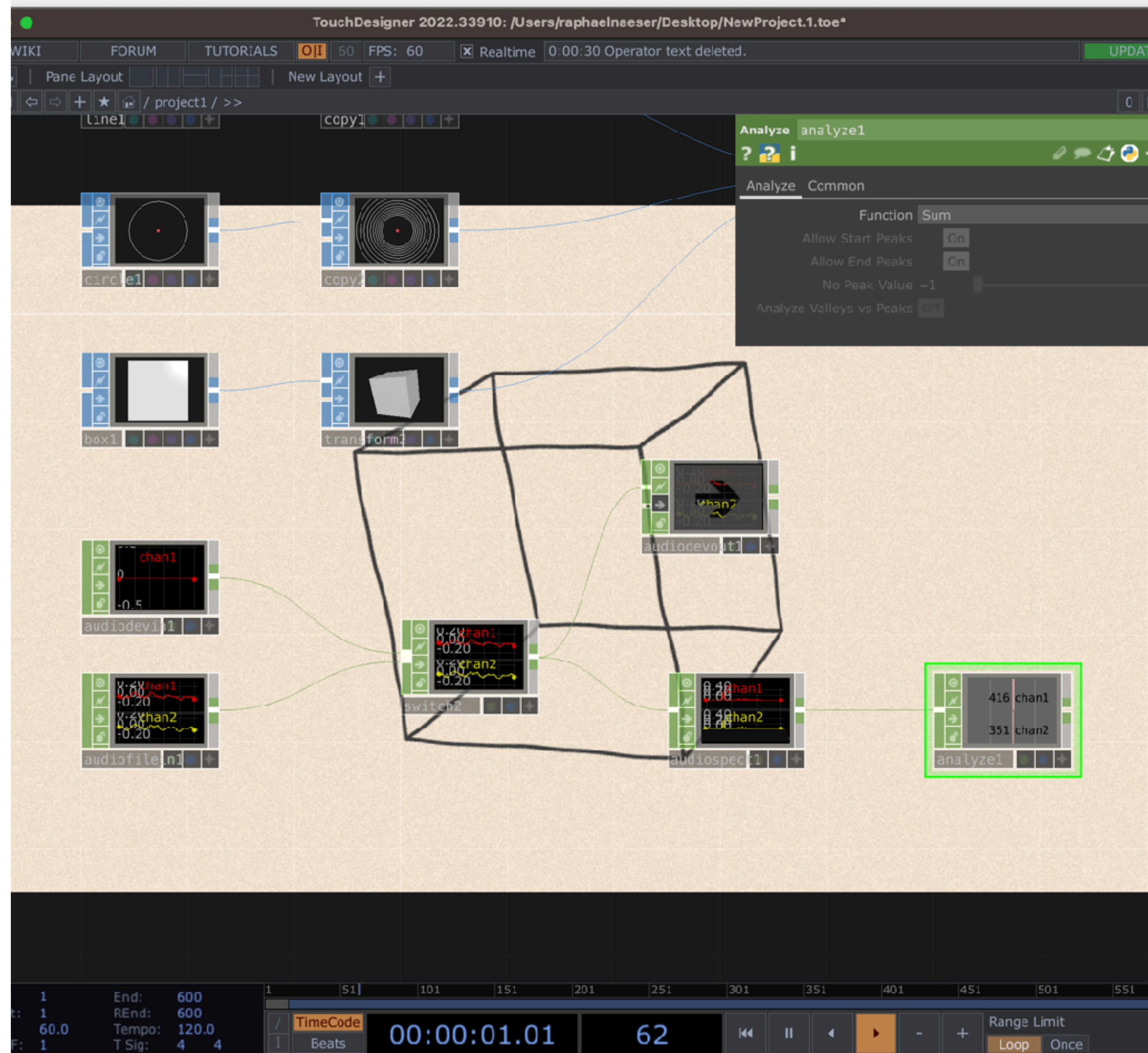
CHOP - Channel Operator



- Rauszoomen
- CHOP Null
 - Umbenennen “sliderout1”
- SOP switch 1
 - Select Input
 - `op('sliderout1')['switch1']`
- Slider1 XY: 10 / 10
- Perform Window zeigen

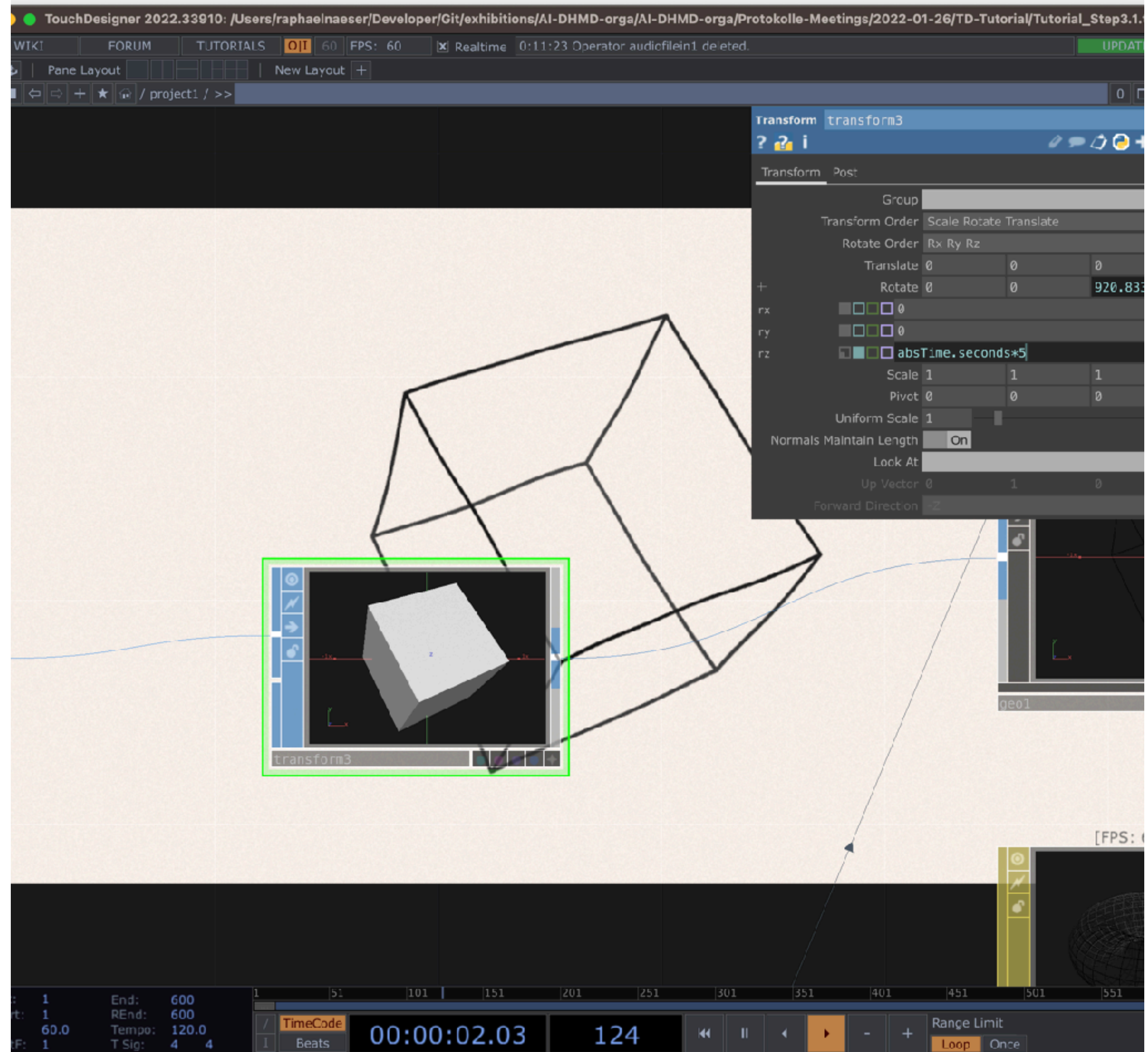


- CHOP Audiodevicein
- CHOP Audiofile
- CHOP Switch
- CHOP AudiodeviceOut
 - Bypass =>
- CHOP Audiodevicein
- CHOP AudioSpectrum
 - FFT 16384
- CHOP Analyze
 - Quantisiert

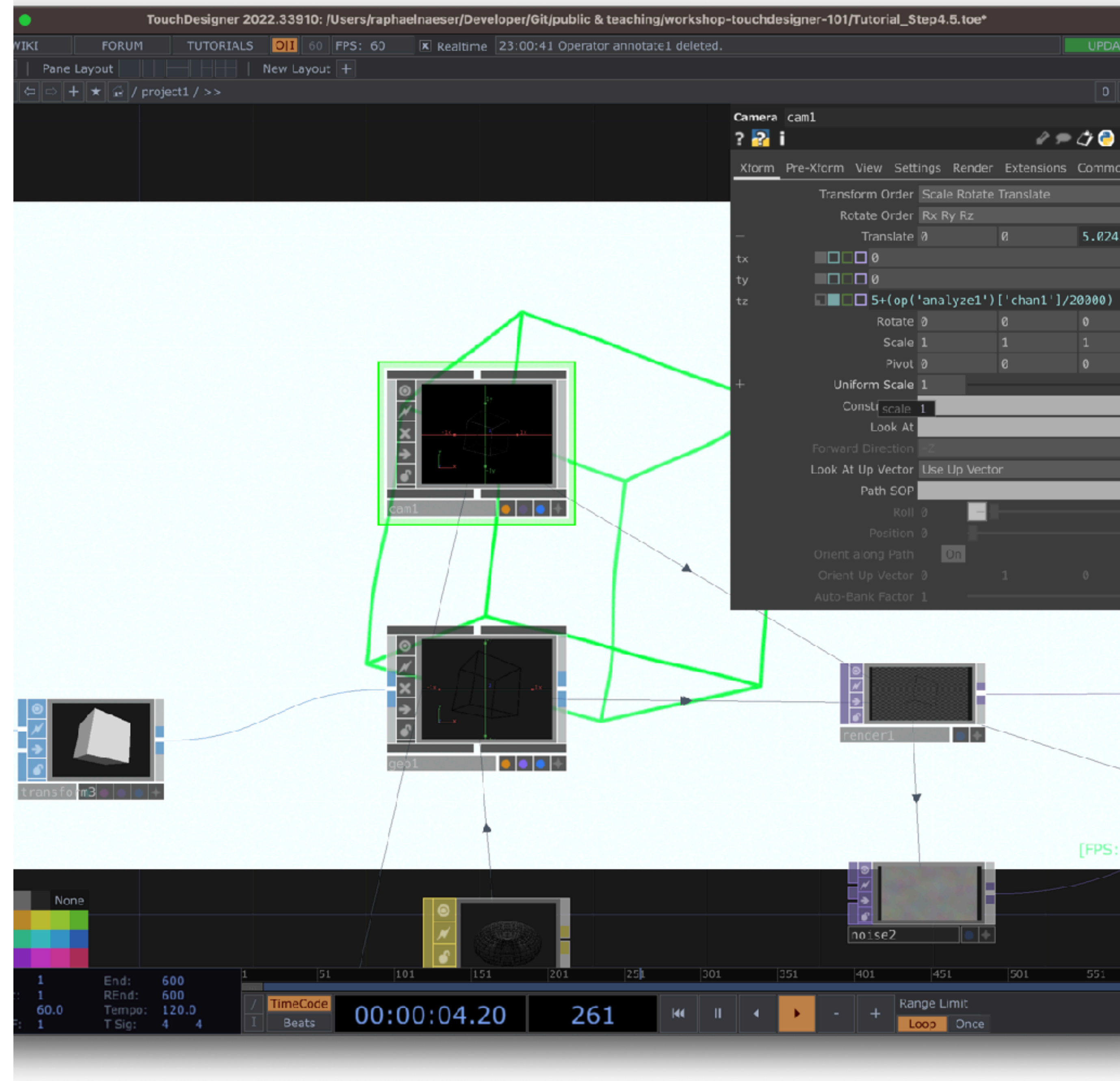


Part 3 - Interaktivität 2

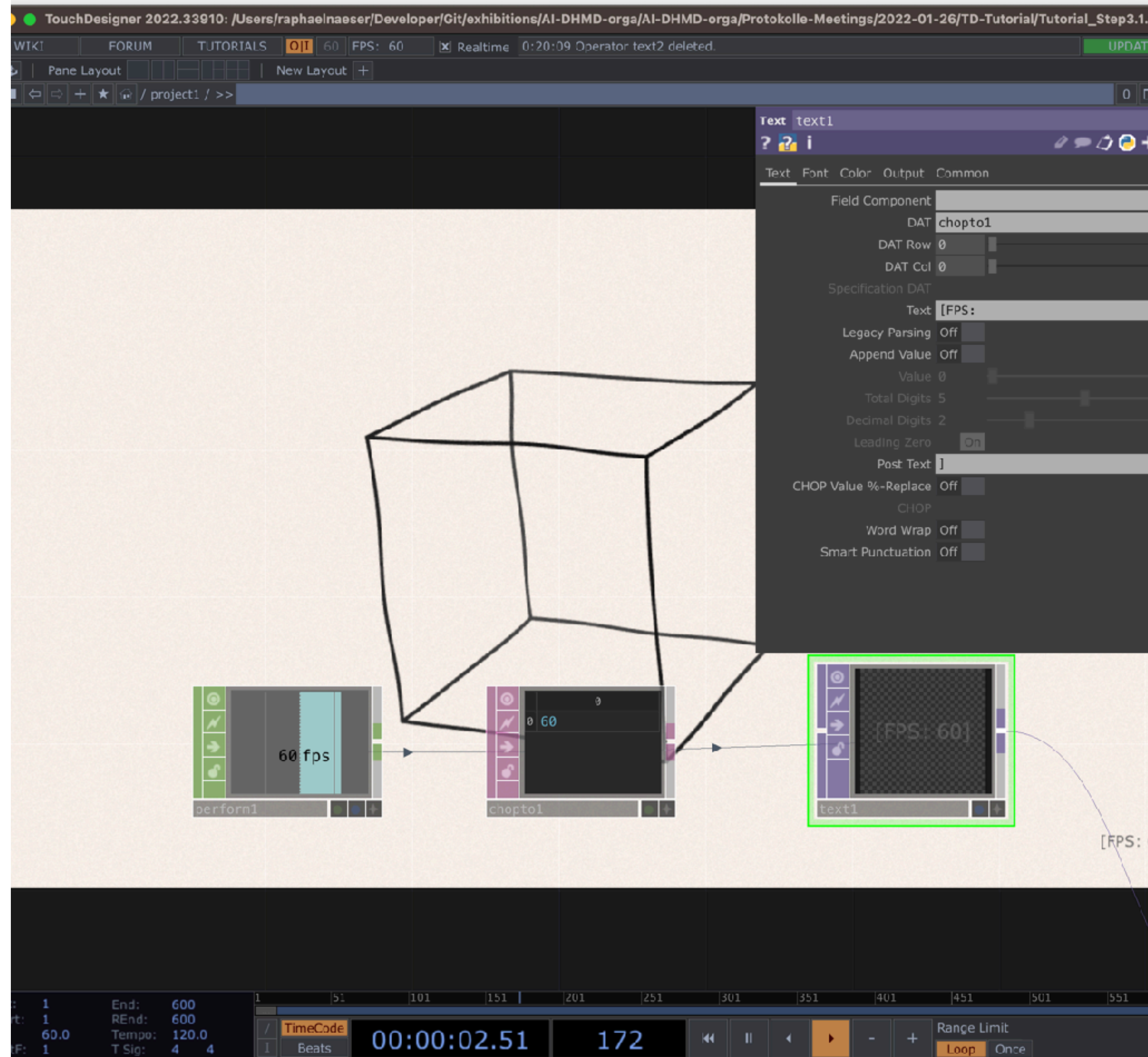
- SOP Transform
- Rz: $\text{absTime.seconds} * 5$



- cam1
- Translate z:
 $5 + (\text{op}(\text{'analyze1'})[\text{'chan1'}] / 20000)$

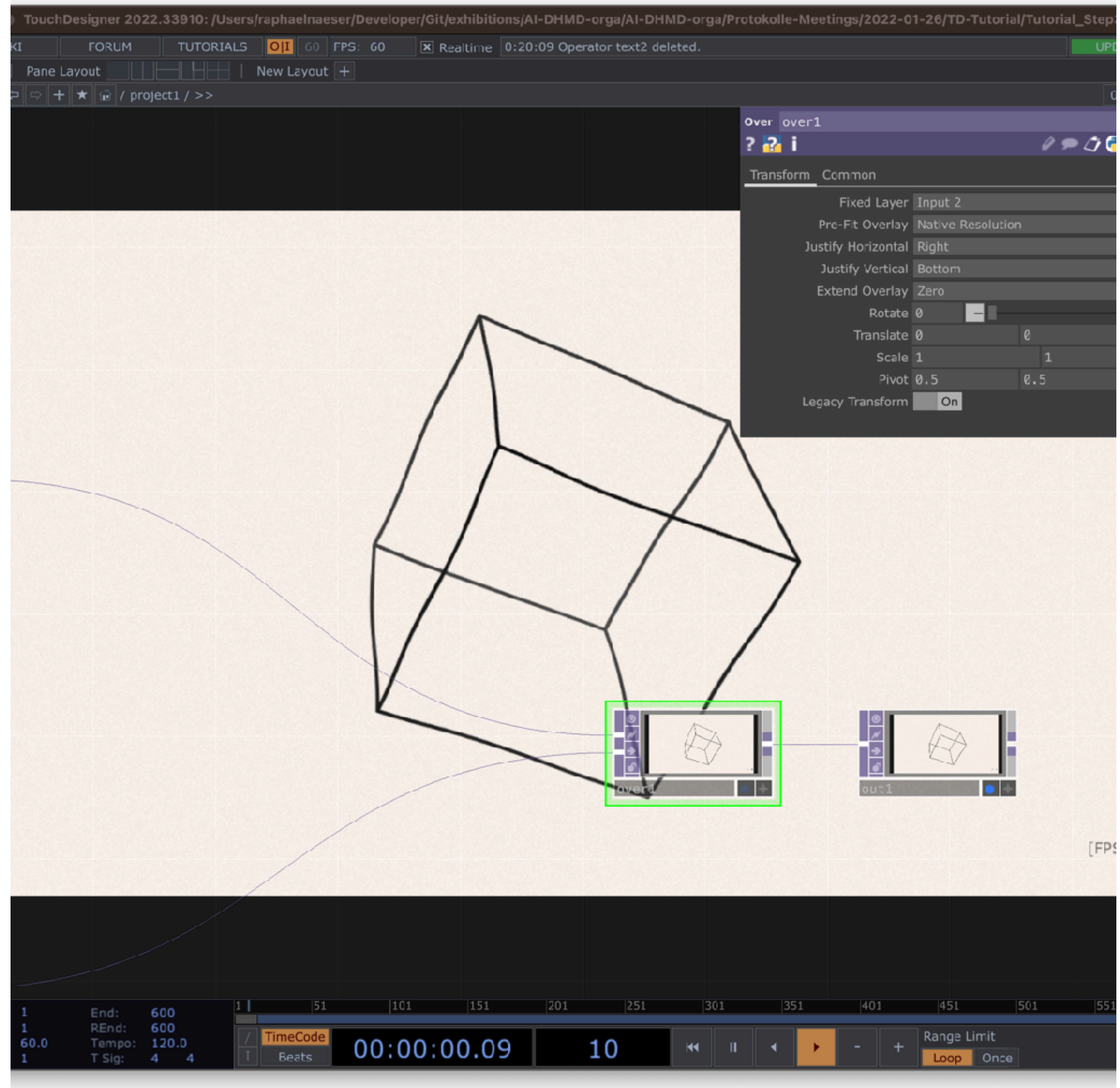


- CHOP Perform
 - Rz: $\text{absTime.seconds} * 5$
- DAT CHOPto
- Perform auf Dat ziehen
- TOP Text
 - “chopto1” in Dat eintragen
 - Text: “[FPS: “
 - Post Text:]
 - Farbe ändern
 - Commons - Resolution 100x100



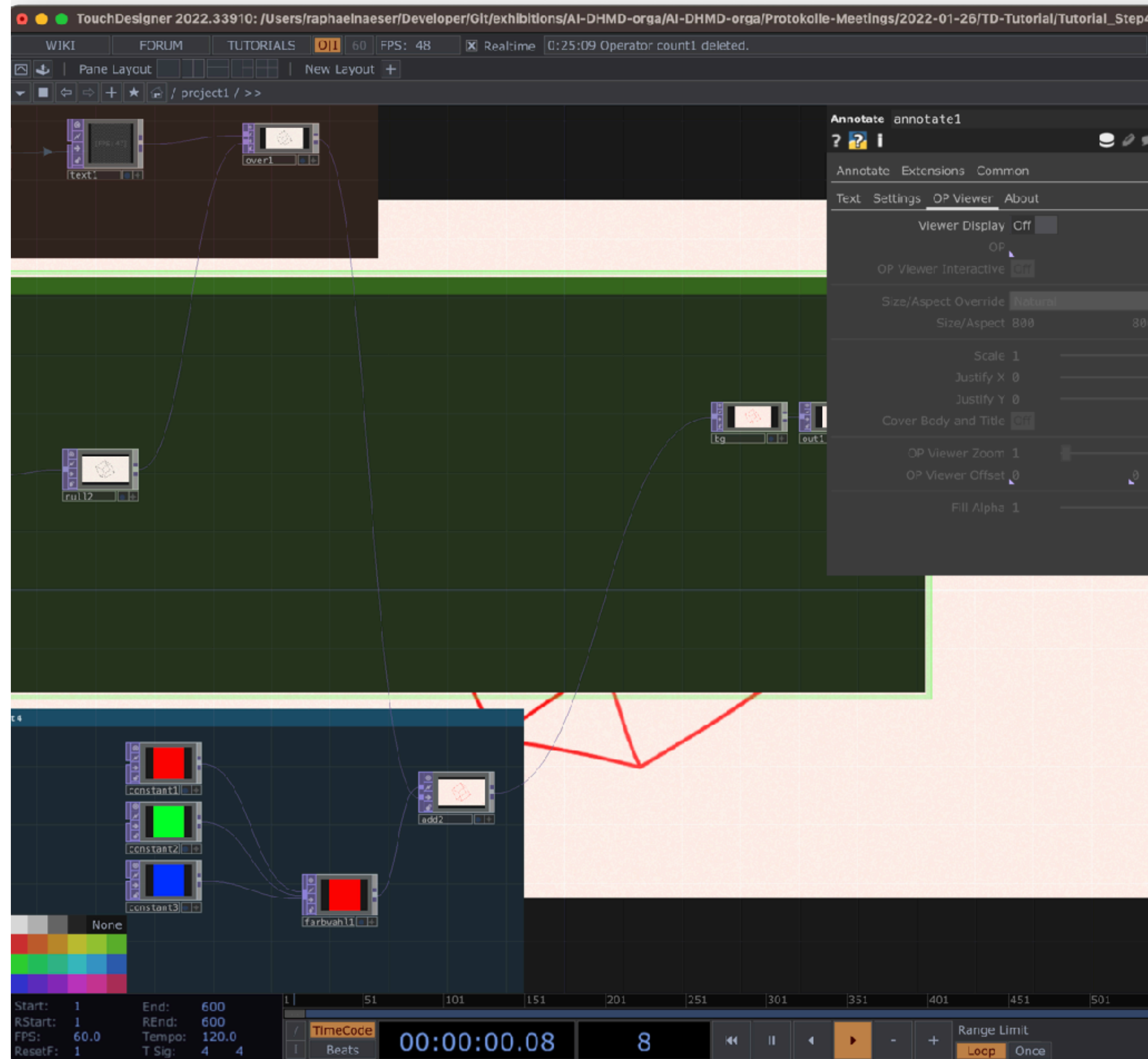
DAT - Daten Operatoren

- TOP Over
 - Unten transform
 - Oben text
 - Profit Overlay: Nat. Res
 - Justify H: Right
 - Justify V: Bottom

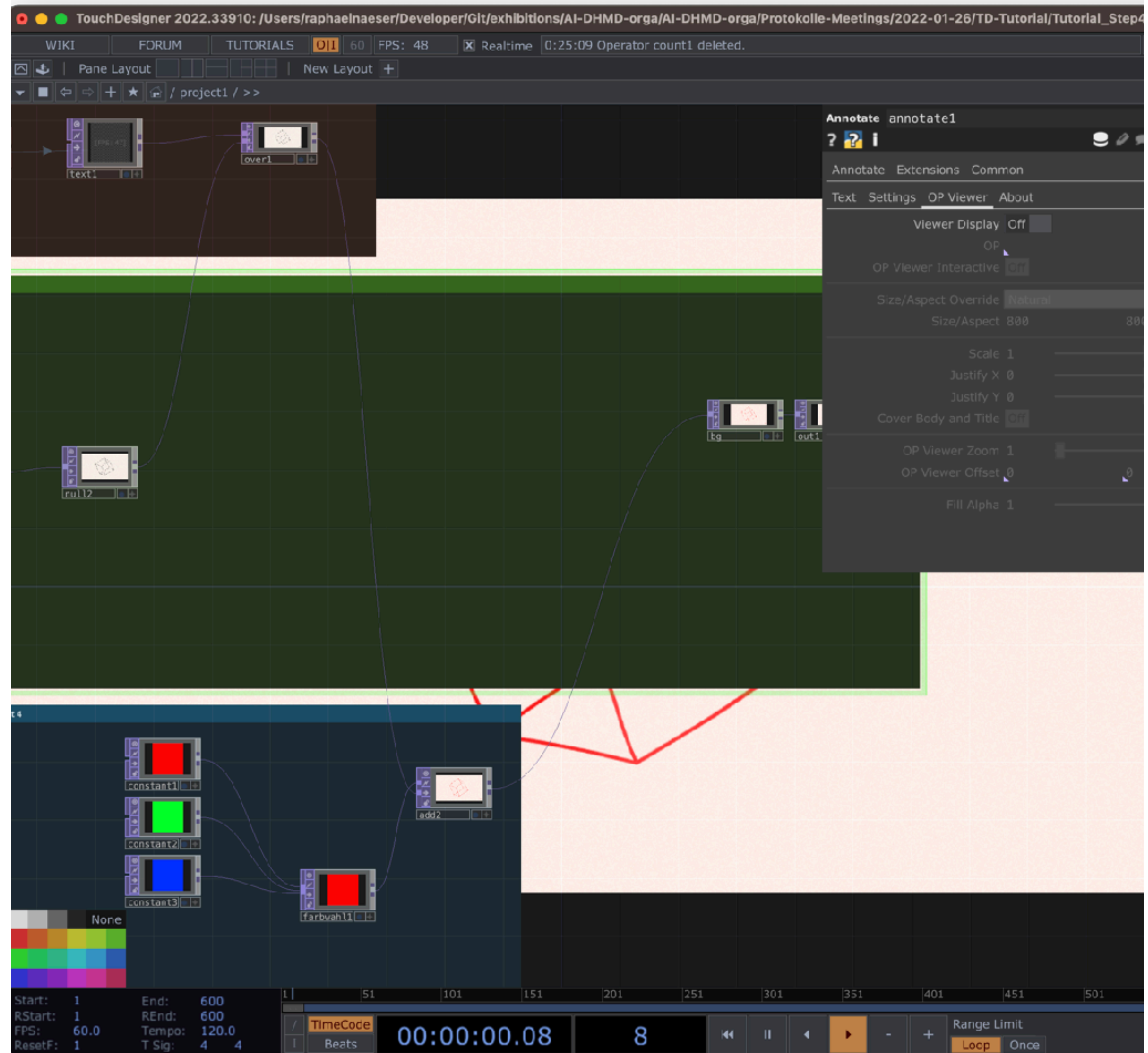


Part 4 - Programmierinterface

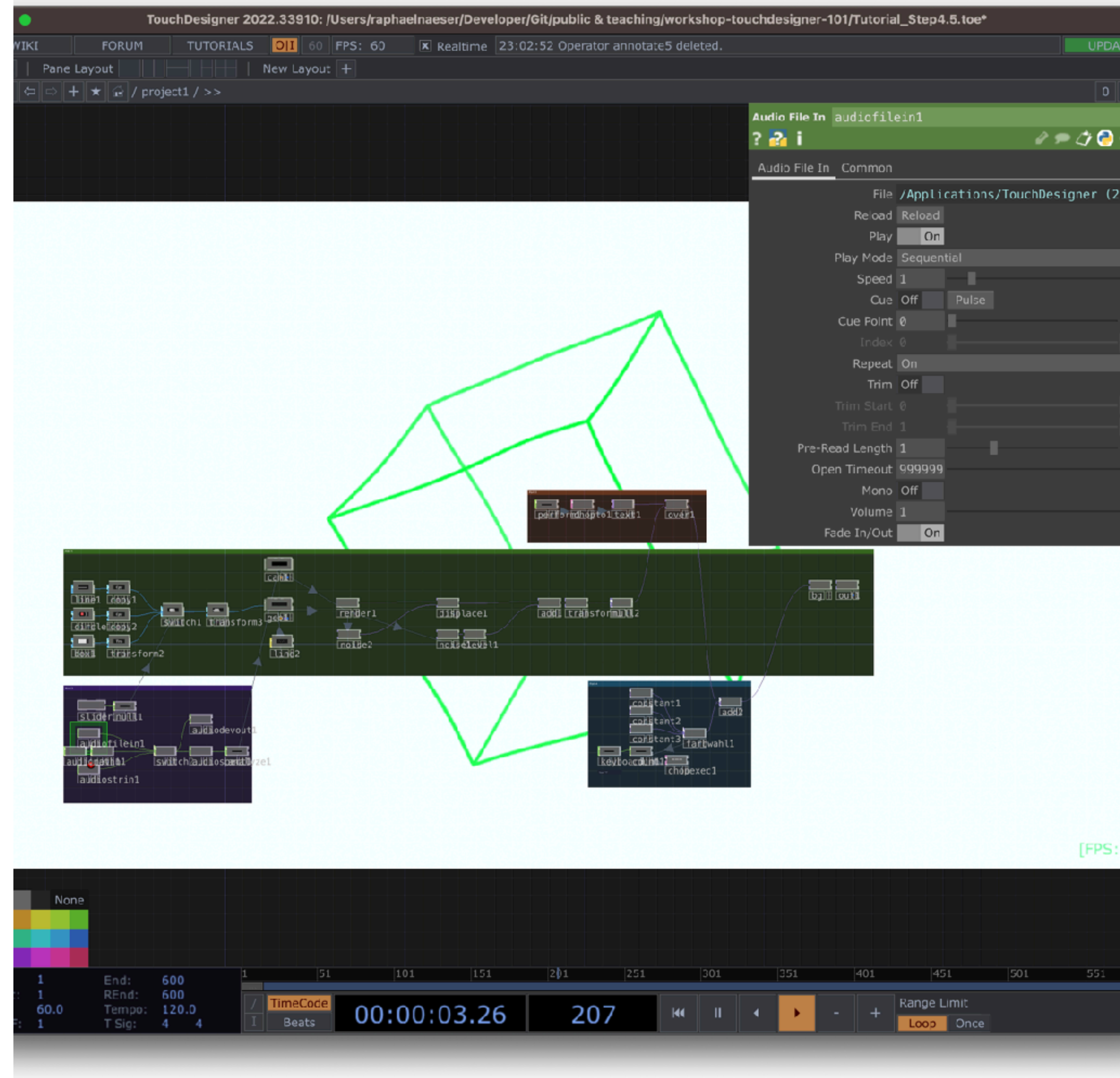
- 3x TOP Constant
 - Unterschiedliche Farben
- TOP Switch “farbwahl1”
- TOP add
 - Over unten
 - farbwahl1 oben



- CHOP Keyboarding
- CHOP Count
 - Limit: Loop Min/Max
 - Limit Maximum:
 - `len(op('farbwahl1').inputs)`
- DAT CHOP Execute
 - Rechtsklick Textport
 - Allen Code bis auf "onValueChanged" löschen
 - Code:
`op('farbwahl1').par.index = val`



- Anmerkungen-, Kommentar-, Network-Boxen mit Rechtsklick
- Kommentare bewegen Todes nicht
- Anmerkungen und Network-Boxen bewegen Codes.
- Können mit dem Node-Farbtool eingefärbt werden. (Alle anderen Codes ebenso)

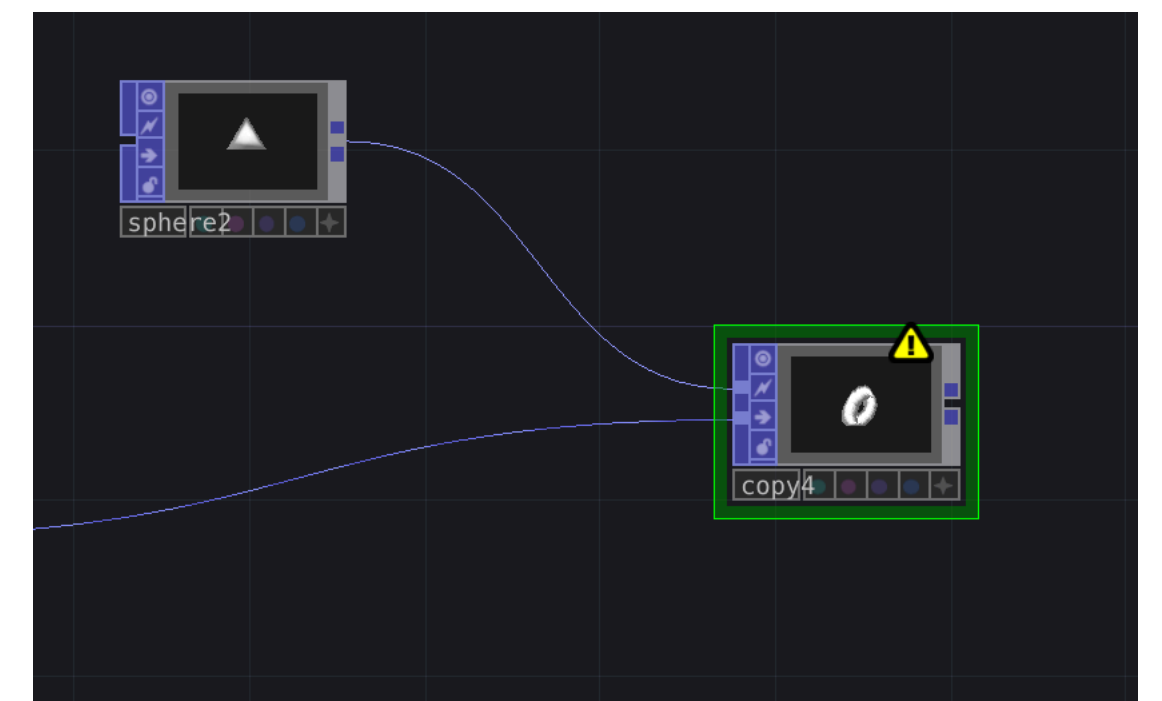


Part 5 - POPs & Feedback

- Point Generator
 - Torus
 - 1000p
 - Orient XY Plane
 - R: 2 / 0.2



- Noise
 - harmonics 0
 - amplitude 0.05
 - transform: Translate 4D absTime.seconds*0.1



- Sphere
 - Tetrahedron

- Copy
 - Template attributes: TempScale: „Scale“



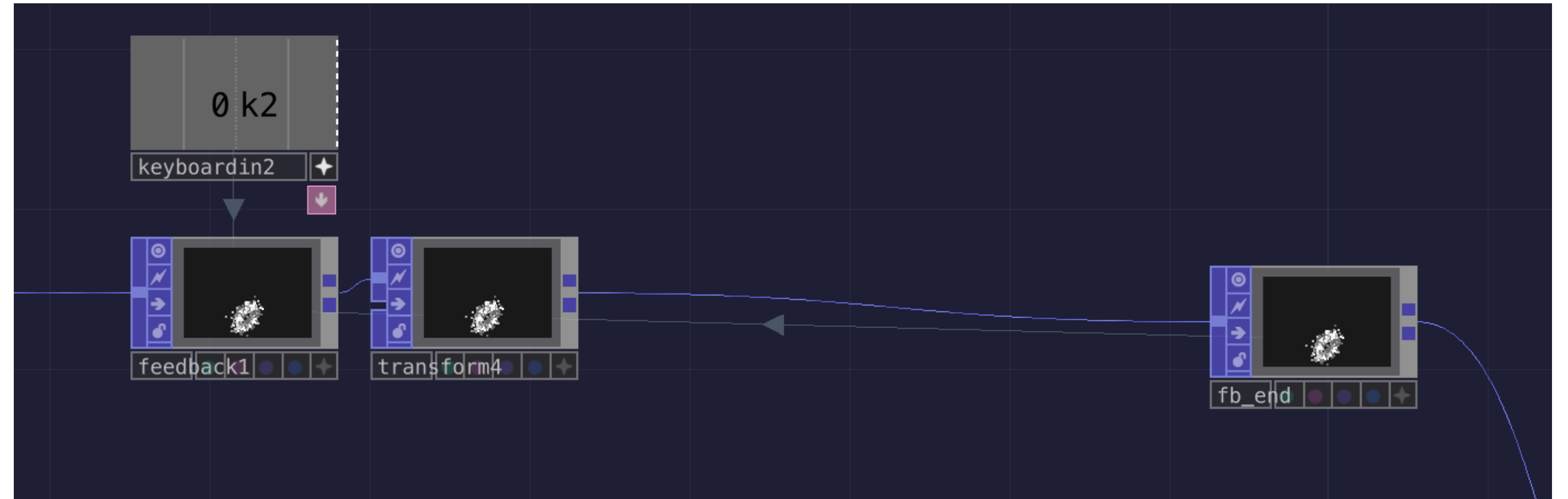
- Noise
 - harmonics 0
 - amplitude 0.5
 - Output -> Noise as Scale

- ReRange
 - Input Attrib: „Scale“
 - Map2low: 0.1
 - Map2High: 5

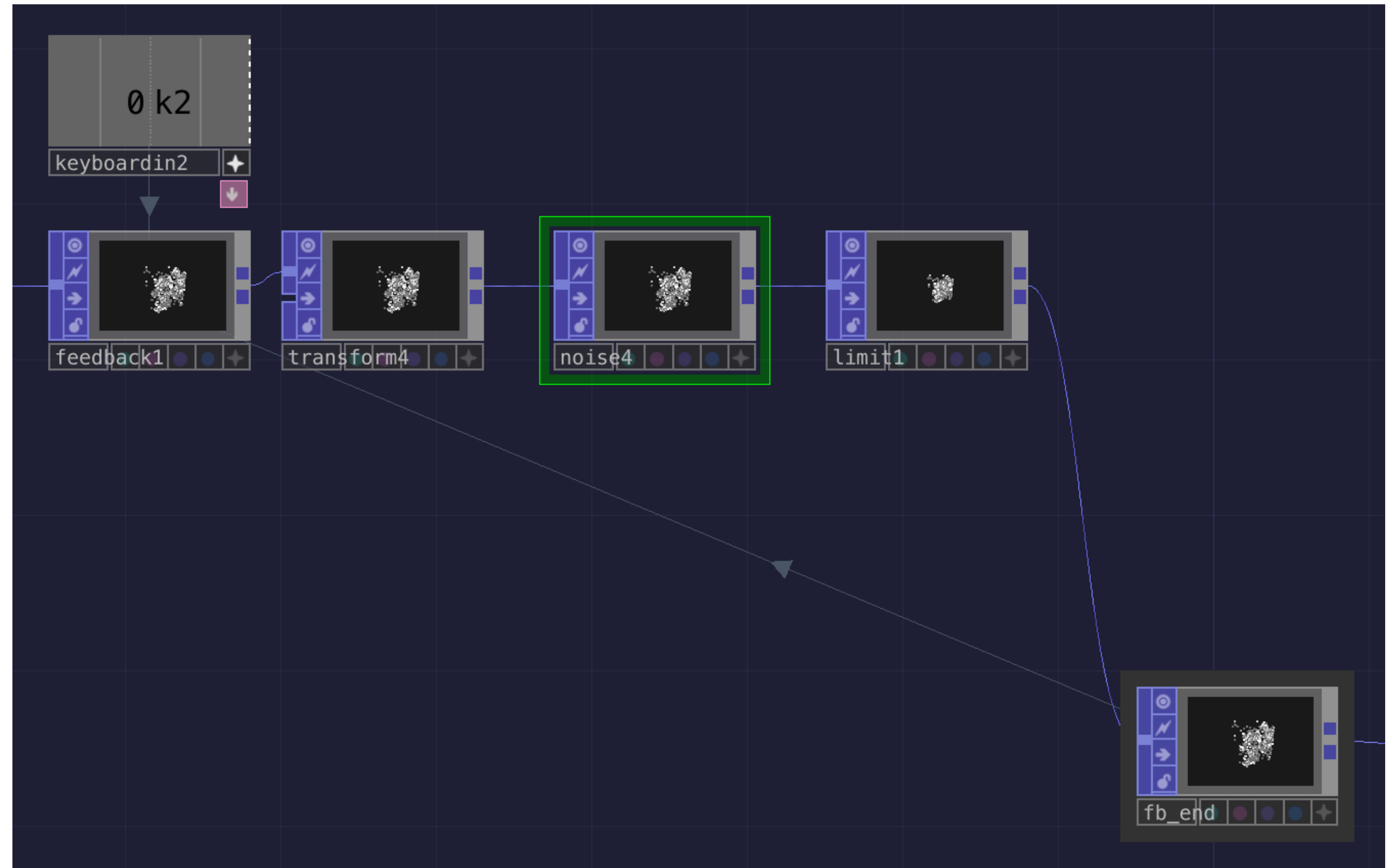
- Proximity Node
 - Max: 0.2
 - Min: 0.01
- Line Mat:
 - Far: Off
 - Color: Black
 - Point multilp 3
- Render on white
- Geos: only take their nodes



- Sprinkle
 - 1000 Points
- Noise
 - Amp 0.7
- Feedback + Keyboard In
 - Keyboard In to chop ref -> Start & Init
- Transform
 - Translate Y -0.01
- Null FB End
 - Drag FB_End to Feedback



- Limit
 - Loop & Loop
 - Min: -2
 - Max: 2
- Noise davor
 - Size: 1
 - Period: 3
 - Harm: 0
 - Hspred: 14.1
 - Hgain: 0.852
 - Amp 0.007



- Comp
- Multiply

